

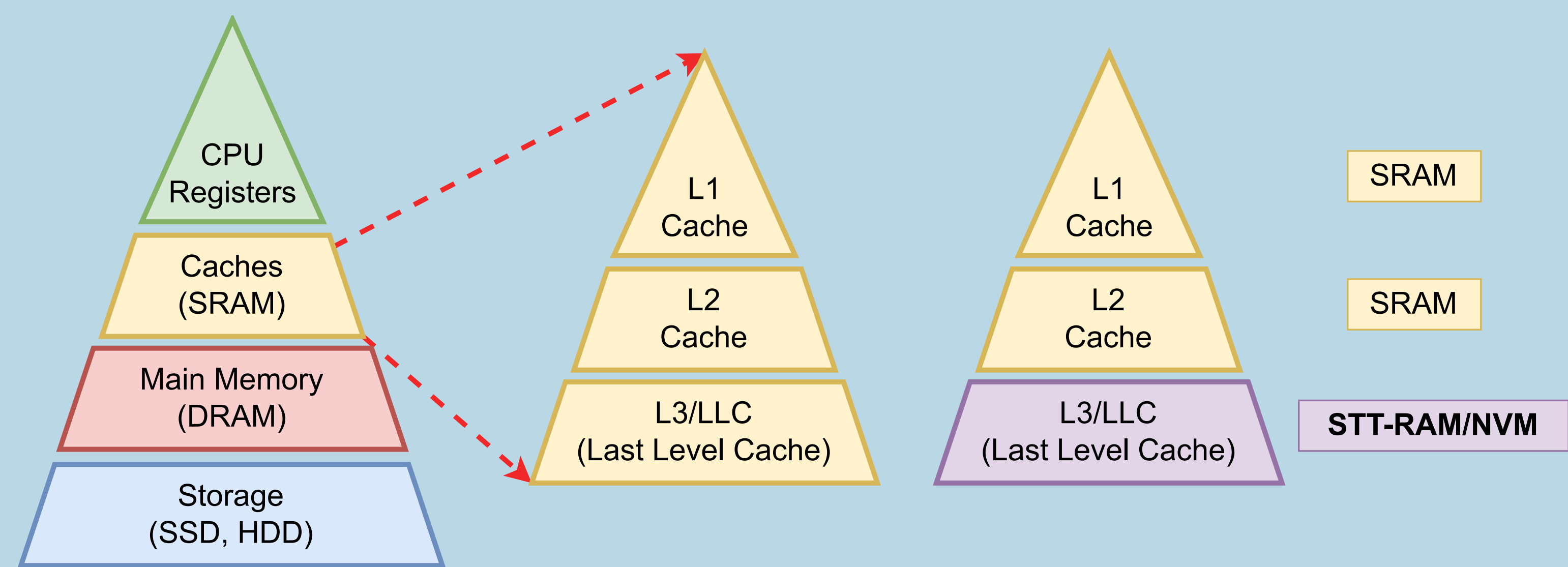
CAMOUFLAGE: An Efficient Mechanism to Hide Congestion in *NVM LLC*

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Introduction

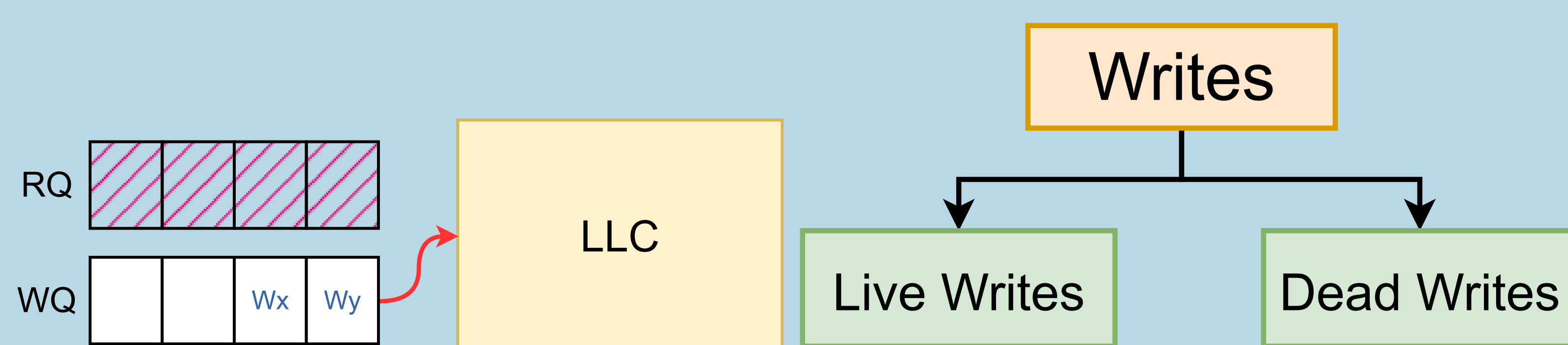
- Increasing core count and expanding workloads necessitate a proportional increase in LLC size.
- Traditional SRAM caches, when enlarged, lead to significant area and power consumption.
- STT-RAM offers high density and low leakage power compared to traditional SRAM.

Memory Hierarchy



Challenges

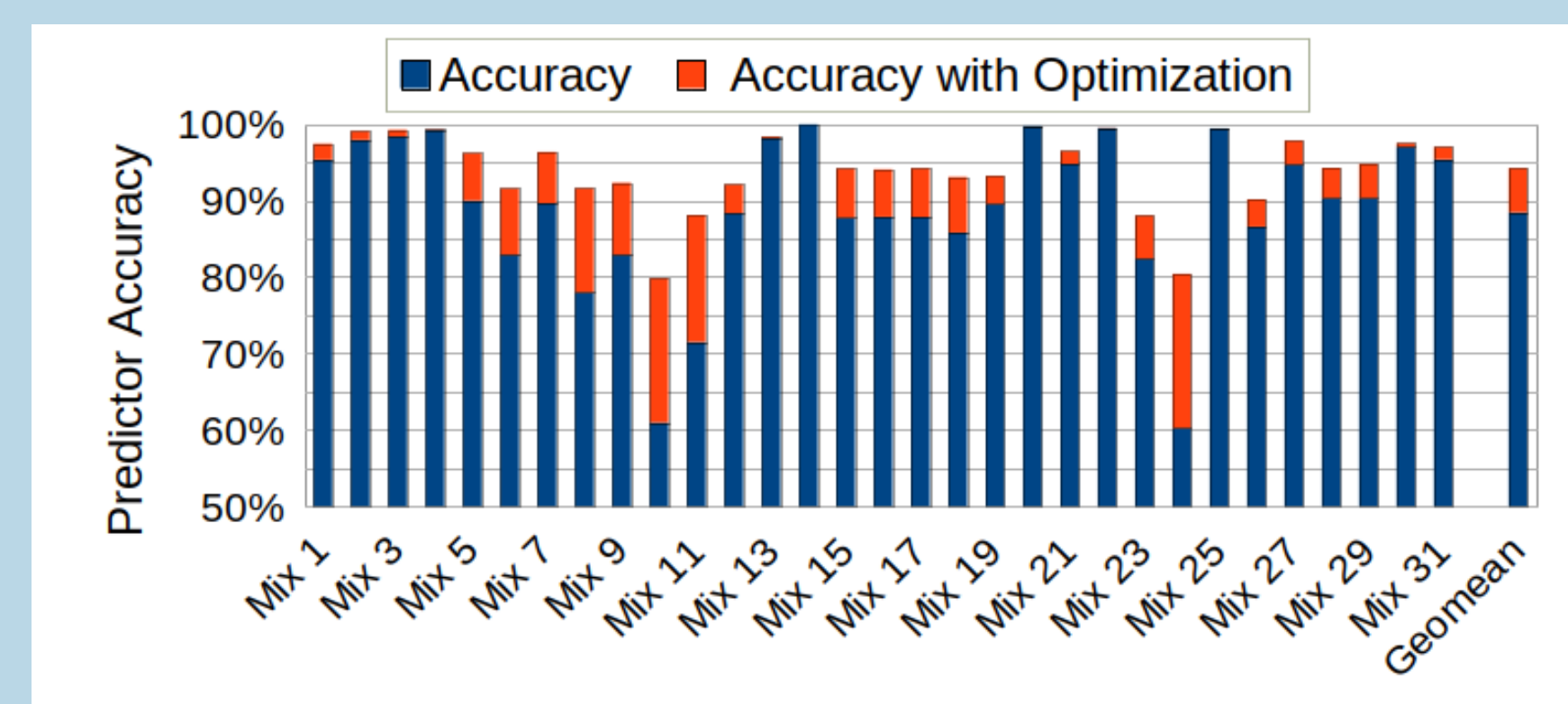
- The read queue experiences **congestion** as STT-RAM LLC is busy performing costly writes.



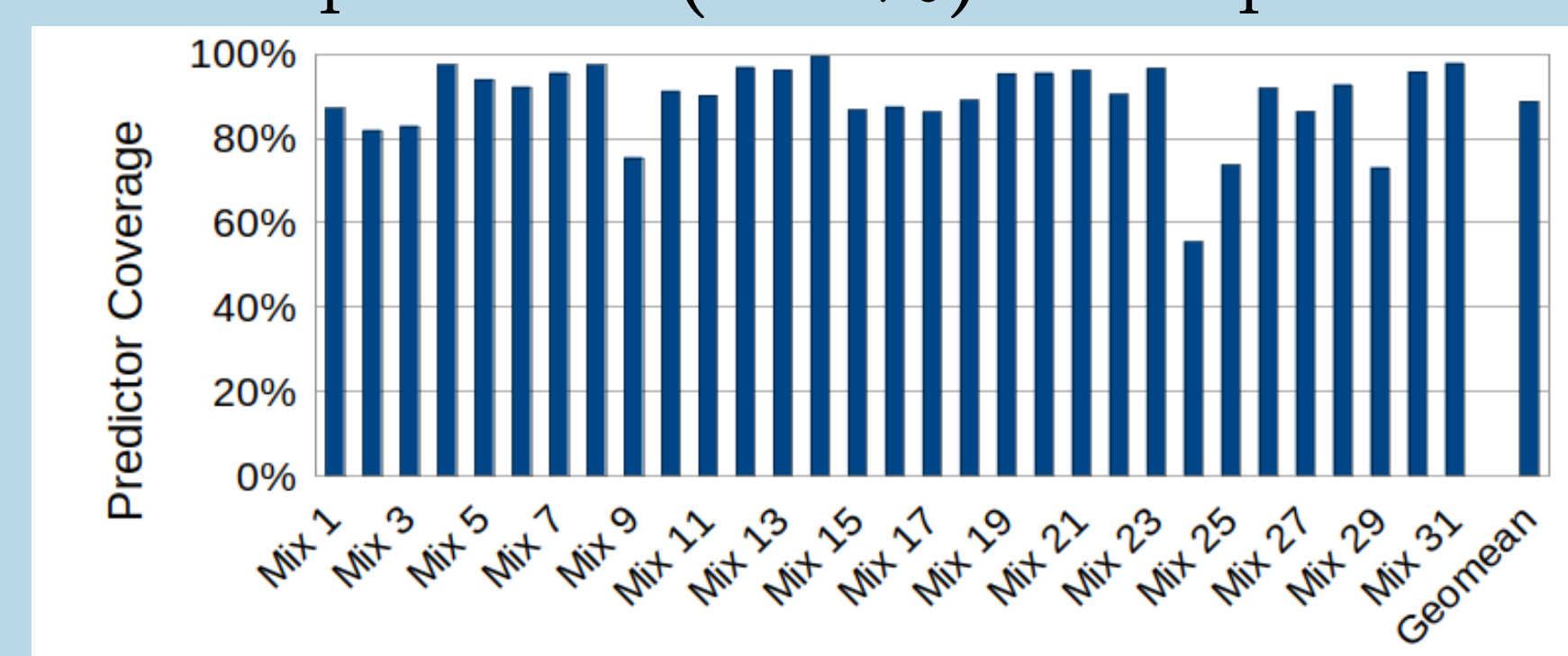
Motivation

- (1) A decade-long emphasis on dead write predictors within the community.
- (2) Limited Attention on impact of Live Writes.

Evaluation



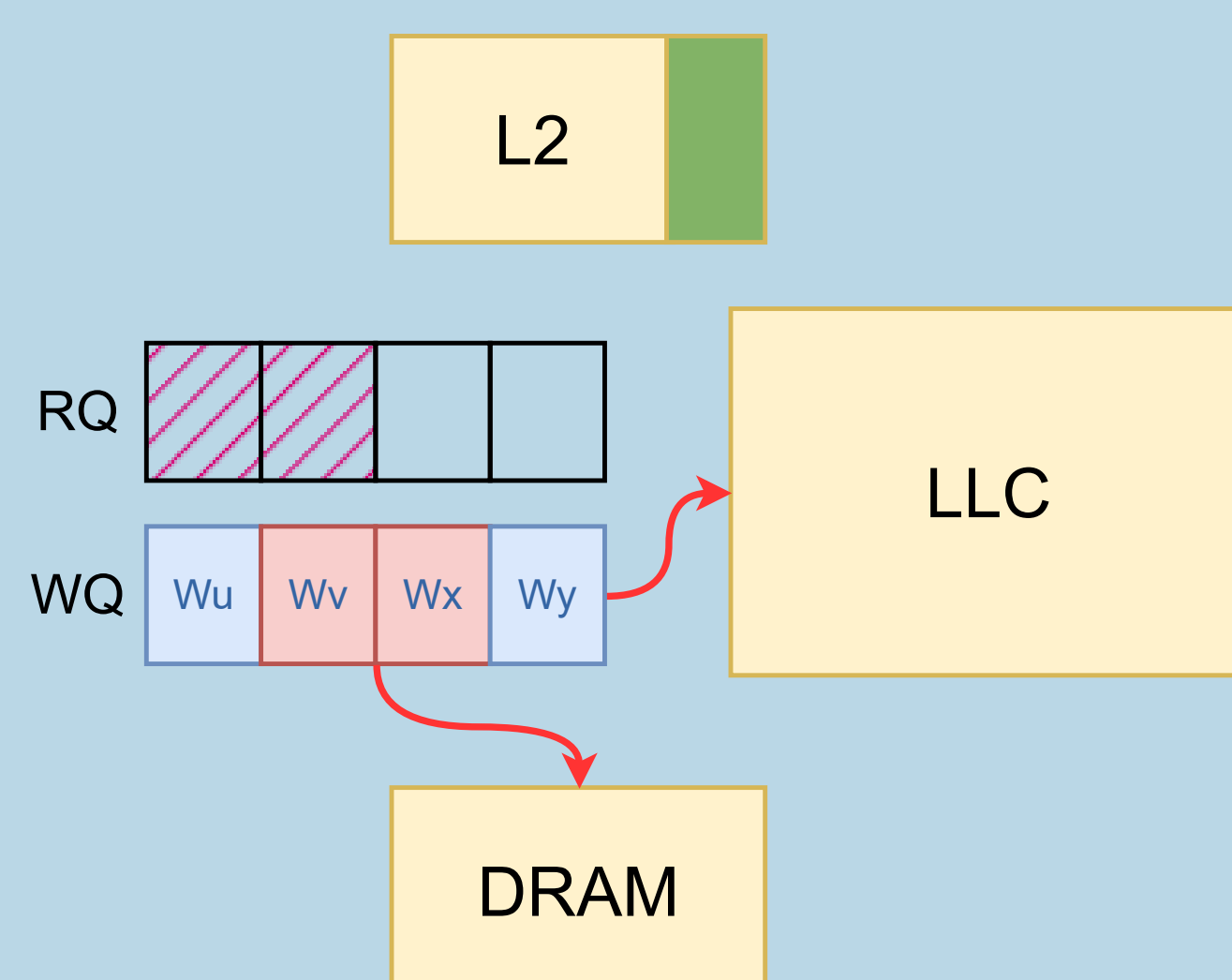
- (1) The **Accuracy** of the predictor is ($\sim 88\%$), further improves to ($\sim 94\%$) with optimization



- (2) The **Coverage** of the predictor is ($\sim 89\%$).

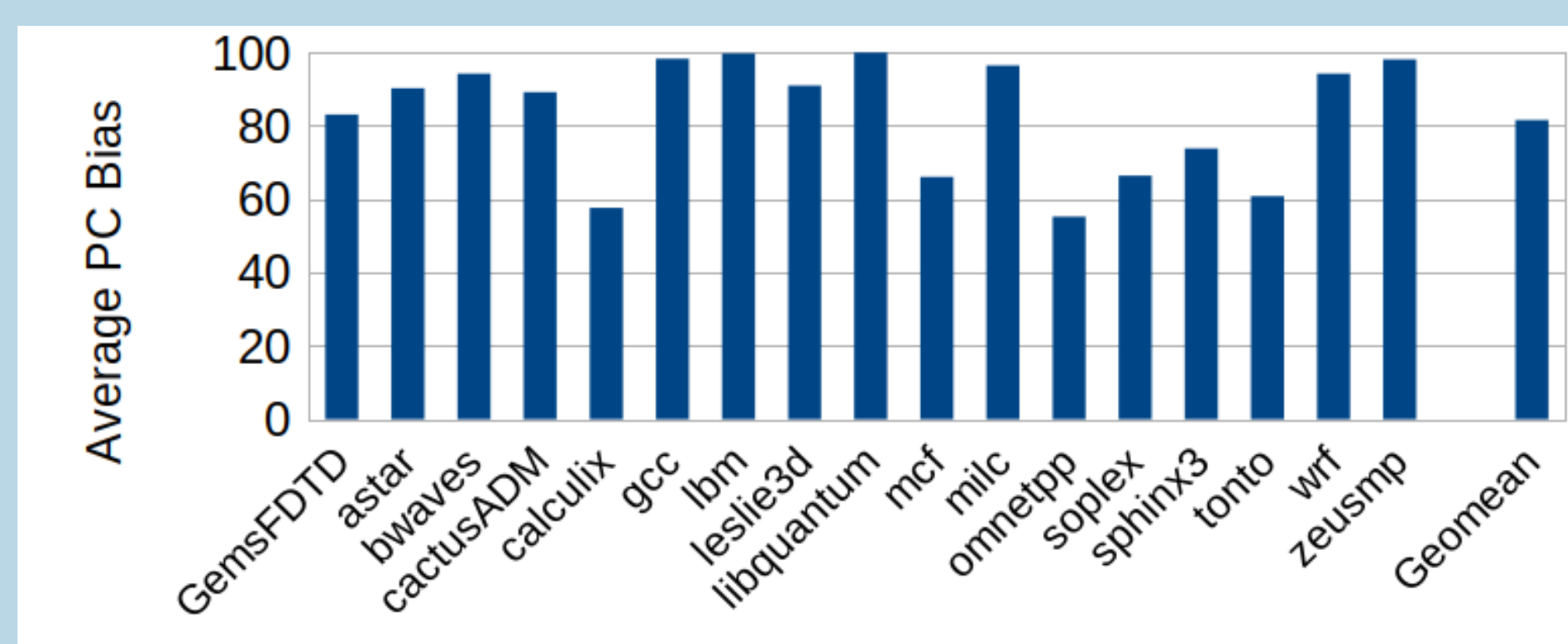
State-of-the-art

- Predicting and bypassing **dead writes** directly to **DRAM** can free up LLC space. Even at the cost of **Low hit rate**.

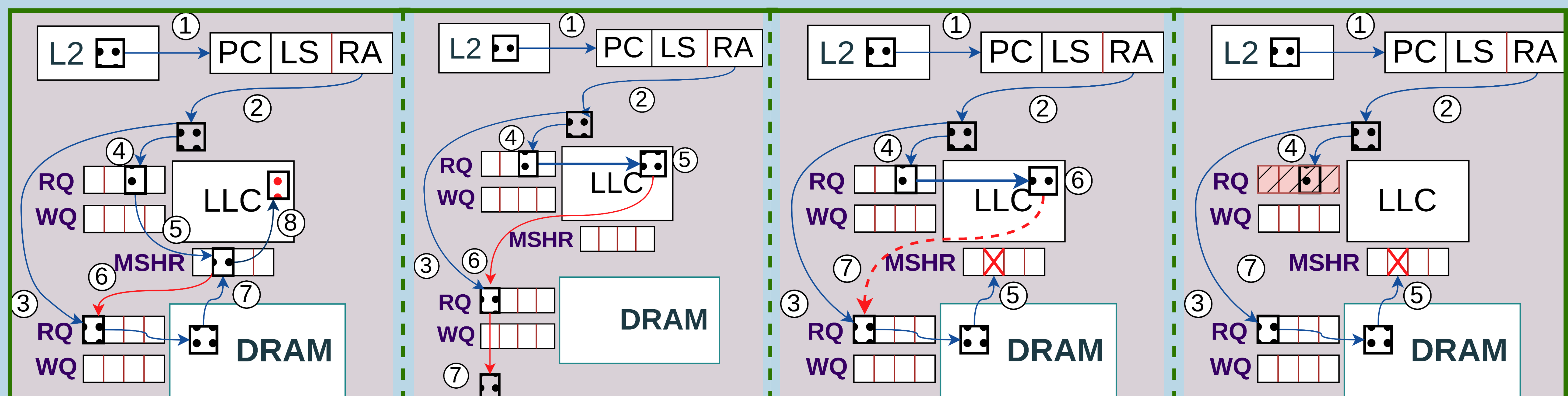


Key Insights

- Questioning **potential** read request **wait** in STT-RAM LLC RQ?
- Correlation with LLC Misses/PC-bias ($\sim 81\%$).



Design of Camouflage



Contributions

- The Read **Run-Ahead (RA)** technique, when used orthogonally with dead write bypassing, effectively hides congestion from **live writes**.
- The RA predictor achieves approximately ($\sim 88\%$) accuracy, which increases to around ($\sim 94\%$) with optimizations, maintaining a coverage of about ($\sim 89\%$).
- The PC-bias and its correlation with L2 and LLC misses for our experiments is ($\sim 81\%$).

Future work

- (1) Coordinating with the prefetch request can further enhance performance.
- (2) Accuracy can be enhanced further by using additional workload attributes alongside the PC.

References

- [1] K. Korgaonkar et al., "Density Tradeoffs of Non-Volatile Memory as a Replacement for SRAM Based Last Level Cache," 2018 ACM/IEEE 45th Annual International Symposium on Computer Architecture (ISCA), Los Angeles, CA, USA, 2018, pp. 315-327, doi: 10.1109/ISCA.2018.00035.

Publication

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